

March 30, 2026

AI-Powered Student Dropout *Prediction System*

Technical Presentation

John Christopher Perez

Technical Content Overview

1. Problem & Data Overview
2. EDA Key Findings
3. Feature Engineering Approach
4. Model Selection & Methodology
5. Results & Model Comparison
6. Confusion Matrix & ROC Curve
7. Feature Importance & Explainability
8. Limitations & Future Work

Predicting student dropout risk using institutional data

- ▶ **Business problem:** institutions often identify at-risk students too late for effective intervention.
- ▶ **ML task:** supervised binary classification.
- ▶ **Target:** 1 = At-Risk (Dropout), 0 = Not At-Risk (Enrolled/Graduate).
- ▶ **Dataset:** 4,424 student records with demographic, academic, financial, and institutional variables.
- ▶ **Primary metrics:** ROC-AUC and Recall for the at-risk class.

4,424

student records

34

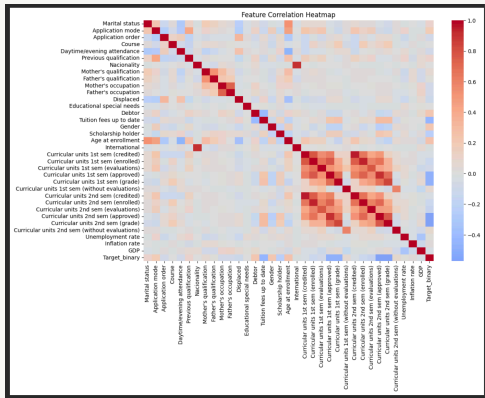
input features

32.1%

at-risk class

EDA KEY FINDINGS

Academic and financial variables show the clearest separation



- ▶ Lower grades and lower approval rates are strongly associated with dropout risk.
- ▶ Students with unpaid tuition or debtor status show higher attrition risk.
- ▶ Older students exhibit moderate risk patterns, but weaker than academic features.
- ▶ EDA suggests that performance, engagement, and financial stability should be prioritized in modeling.

We engineered features to capture progress, consistency, and engagement

Engineered Features

- ▶ Total approved units
- ▶ Total enrolled units
- ▶ Total evaluated units
- ▶ Average grade across semesters
- ▶ Approval rate (1st semester)
- ▶ Approval rate (2nd semester)
- ▶ Overall approval rate

Preprocessing Pipeline

- ▶ One-hot encoding for categorical variables
- ▶ Standard scaling for numerical variables
- ▶ Binary variables passed through directly
- ▶ Stratified train-test split

Multiple classifiers were evaluated using a consistent pipeline

Models Compared

- ▶ Logistic Regression
- ▶ Decision Tree
- ▶ Random Forest
- ▶ Support Vector Machine
- ▶ XGBoost

Evaluation Logic

- ▶ Same preprocessing for all models
- ▶ Binary classification setting
- ▶ Compare Accuracy, Precision, Recall, F1, ROC-AUC
- ▶ Prioritize **Recall** and **ROC-AUC** for early warning use case

RESULTS & MODEL COMPARISON

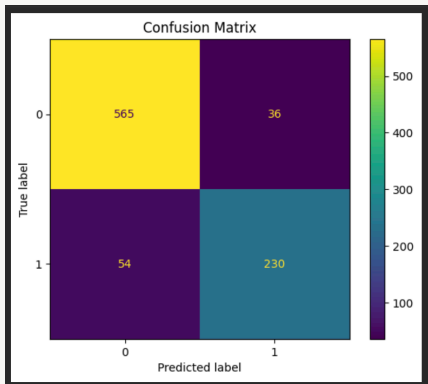
XGBoost delivered the best balance of detection and overall performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.884	0.872	0.746	0.805	0.934
Logistic Regression	0.888	0.874	0.761	0.814	0.931
SVM	0.882	0.905	0.708	0.794	0.927
XGBoost	0.898	0.865	0.810	0.836	0.926
Decision Tree	0.818	0.711	0.729	0.720	0.795

Model choice: XGBoost was selected as the final model because it achieved the highest Recall and F1-score, making it the strongest option for identifying at-risk students early.

FINAL MODEL EVALUATION

Confusion matrix: strong detection of at-risk students

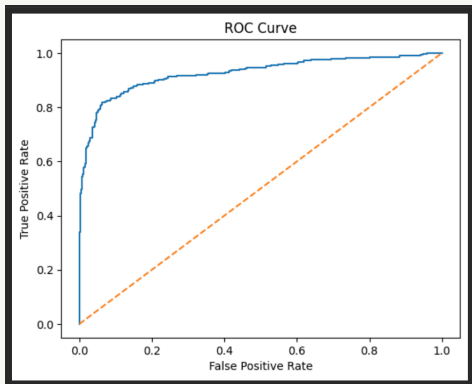


- ▶ True Negatives: 565
- ▶ True Positives: 230
- ▶ False Positives: 36
- ▶ False Negatives: 54

The model correctly identifies most at-risk students while keeping false alarms relatively low. In an educational setting, this trade-off is acceptable because missing an at-risk student is more costly than unnecessary intervention.

FINAL MODEL EVALUATION

ROC curve: excellent class discrimination

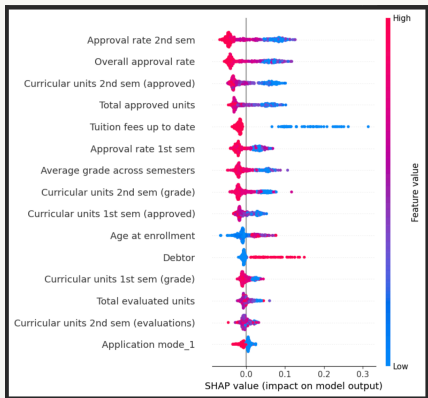


- ▶ ROC-AUC = 0.926
- ▶ Strong upward curve toward the top-left corner
- ▶ High true positive rate at relatively low false positive rates

This indicates that the model can reliably distinguish at-risk students from not at-risk students across different classification thresholds.

FEATURE IMPORTANCE & EXPLAINABILITY

SHAP confirms that academic performance and finances drive risk



- Low approval rates increase dropout risk
- Lower approved units push predictions toward at-risk
- Tuition not up to date and debtor status are strong risk signals
- Age has moderate influence relative to performance variables

LIMITATIONS & FUTURE WORK

Key constraints and next technical improvements

Limitations

- ▶ Moderate class imbalance
- ▶ Limited behavioral and psychological variables
- ▶ May not generalize fully across institutions
- ▶ Financial variables may reflect structural inequalities

Future Work

- ▶ Add behavioral engagement data
- ▶ Test external datasets from other institutions
- ▶ Deploy model in a dashboard or LMS workflow
- ▶ Monitor fairness and recalibrate thresholds over time

A close-up, slightly blurred photograph of a dark, textured chalkboard. Two pieces of chalk, one light green and one light yellow, are resting on the chalkboard's ledge. The text "Thank you!" is written in a white, serif font on the right side of the board.

Thank you!

John Christopher Perez